13 If Eddington's name sounds familiar, it might be because he made an eclipse expedition in 1919 that provided some of the first observational evidence for Einstein's general theory of relativity. The observation of stars whose light grazed past the Sun en route to us showed that the light was being bent by the Sun's distortion of space. (This is the kind of observation that can only happen when the Sun is eclipsed.) A famous headline at the time proclaimed, "LIGHTS ALL ASKEW IN THE HEAVENS—MEN OF SCIENCE MORE OR LESS AGOG OVER RESULTS OF ECLIPSE OBSERVATIONS." Women of science were, presumably, unimpressed.

14 This would be an excellent pun if Type Ia were likely to be able to create gold. While they can create other elements during the explosion (impressive amounts of nickel, for instance), due to the extreme temperatures and pressures involved, gold is probably mostly made in collisions of neutron stars. Alas.

6. Vacuum Decay

1 Quarks come in six different "flavors," which have different masses and charges. The flavors are: up, down, top, bottom, charm, and strange. They were named in the 1960s.

2 In Kurt Vonnegut's *Cat's Cradle*, a new form of ice is created, "ice-nine," that is more stable than liquid water. In the story, every bit of water a particle of ice-nine touches turns to ice-nine, creating an existential threat to life and the world.

3 Believe me, I know.

4 This is due to something called the Askaryan Effect, in which an ultra-high-energy neutrino punches through the lunar regolith and creates a burst of radio waves that we can hopefully pick up with radio telescopes. So far, our telescopes haven't been sensitive enough, but we should be able to pick up these signals with the next generation of instruments.

5 This part of the book was written in North Carolina in August.